

Lincoln University College
Faculty of Computer Science and Multimedia

Roadside Assistance System
A PROJECT PROPOSAL

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1 Introduction

Roadside emergencies such as tire punctures, battery failure, and minor mechanical issues are common challenges faced by travelers in Nepal, especially in rural and highway areas where organized support is limited. Most people still depend on personal contacts or local workshops, which often leads to delays, inconvenience, and safety risks. Although a few applications like Mobile Garage and Road Sewa exist, their coverage, adoption, and real-time availability features remain limited, leaving many regions underserved.

To address these gaps, this project proposes a **Roadside Assistance System**, a simple digital platform that connects stranded users with nearby available mechanics or helpers. The system allows users to create service requests, view available helpers, track request status, and provide feedback. Mechanics can register themselves, update availability, and accept service requests. A basic location-based matching algorithm will be implemented to recommend the nearest helper.

2 Problem Statement

Despite the growing number of vehicles in Nepal, there is no reliable digital system that connects stranded travelers with nearby mechanics or roadside assistance providers. When issues such as flat tires, battery failure, or minor mechanical problems occur, especially in highways, rural areas, or unfamiliar locations, travelers often face significant delays, safety risks, and difficulty in locating help. At the same time, many skilled local mechanics in different regions remain underutilized due to the lack of an organized platform where they can offer their services. The absence of a centralized, real-time assistance system creates inefficiency and inconvenience for both travelers and service providers. Therefore, there is a need for a platform that can link users with nearby available helpers quickly, efficiently, and transparently.

3 Objectives

The objectives of this project are:

- To develop a platform that allows users to request roadside assistance for common vehicle issues.
- To enable helpers to register, manage availability, and receive nearby service requests.
- To implement a simple algorithm to identify the nearest available helper.
- To manage service requests from creation to completion while supporting local employment.

4 Methodology

4.1 Requirement Identification

4.1.1 Study of Existing System

Currently, roadside assistance services in Nepal are limited and largely unorganized. When a vehicle experiences issues such as a flat tire, dead battery, or minor mechanical faults, travelers often rely on personal contacts, local workshops, or bystanders for help. This approach is time-consuming, uncertain, and especially challenging in rural or unfamiliar areas.

In recent years, applications like **Mobile Garage** and **Road Sewa** have begun providing digital roadside assistance services in Nepal. These platforms allow users to request help and connect with registered mechanics. While they represent an initial effort to digitize roadside services, their adoption remains limited, and many rural regions are still underserved. Furthermore, these existing applications often have restricted functionality in areas such as real-time availability of mechanics, open registration for local service providers, and coverage of diverse minor vehicle issues.

Overall, the current landscape demonstrates the potential of digital roadside assistance in Nepal but also highlights significant gaps. There is a clear need for a more accessible, user-friendly, and regionally adaptable system that efficiently connects stranded travelers with nearby mechanics. This system should prioritize open registration for local helpers, real-time matching, and broader coverage, particularly in semi-urban and rural areas.

4.1.2 Literature Review

Recent studies have explored different approaches to delivering roadside assistance using mobile and web technologies. Research such as “On Road Vehicle Breakdown Help Assistance Web Application” (IJSRST, 2024) and “Roadside Assistance for Vehicle Breakdown” (IJRASET, 2024) highlights the importance of real-time location tracking, user–mechanic interaction, and service request management. These studies demonstrate that integrating GPS, mobile connectivity, and structured workflows can significantly improve response time and service reliability.

Other works like “Emergency Vehicle Breakdown Assistance using Android” (2022) emphasize the role of mobile-based systems for connecting users with nearby mechanics, suggesting simplified user interfaces and quick request handling.

International models such as AAA roadside services in the United States illustrate the potential scalability of such systems. However, these implementations are not adapted to Nepal’s road conditions, connectivity issues, and mechanic availability.

The literature shows increasing use of digital roadside assistance systems, while highlighting a lack of solutions tailored to Nepal’s geography and service accessibility.

4.1.3 Requirement Analysis

Functional Requirements:

1. User registration, login, and profile management
2. Helper registration and availability management
3. Creation and tracking of service requests
4. View nearby helpers
5. Helper request acceptance
6. Service status updates
7. Search/filter by issue type
8. Basic notifications
9. Admin monitoring module

Non-Functional Requirements

1. Usability: Simple and easy-to-use UI
2. Performance: Quick response for CRUD actions
3. Scalability: Ability to add more helpers and users
4. Reliability: Data must remain consistent
5. Security: Authentication and role-based access
6. Availability: System should be accessible whenever needed

4.2 Feasibility Study

4.2.1 Technical Feasibility

The project uses readily available technologies such as HTML/CSS/JS/React/Dart (Not sure) for frontend and Node and firebase for backend. CRUD operations are simple and supported by all modern frameworks. The nearest-helper algorithm is mathematically simple. Hence, the project is technically feasible.

4.2.2 Operational Feasibility

Users are familiar with mobile apps and web systems. Mechanics can easily update availability and accept requests. The flow is simple, requiring minimal training. Therefore, operational feasibility is high.

4.2.3 Economic Feasibility

Development tools (VS Code, MySQL, Node, Python, etc.) are free. No special hardware is required. Hosting can be done on low-cost shared servers or locally. Thus, the project is economically feasible.

4.3 High Level Design of System

The flowcharts show how users request road assistance and track service, while helpers receive requests and update completion status, illustrating the system's overall workflow.

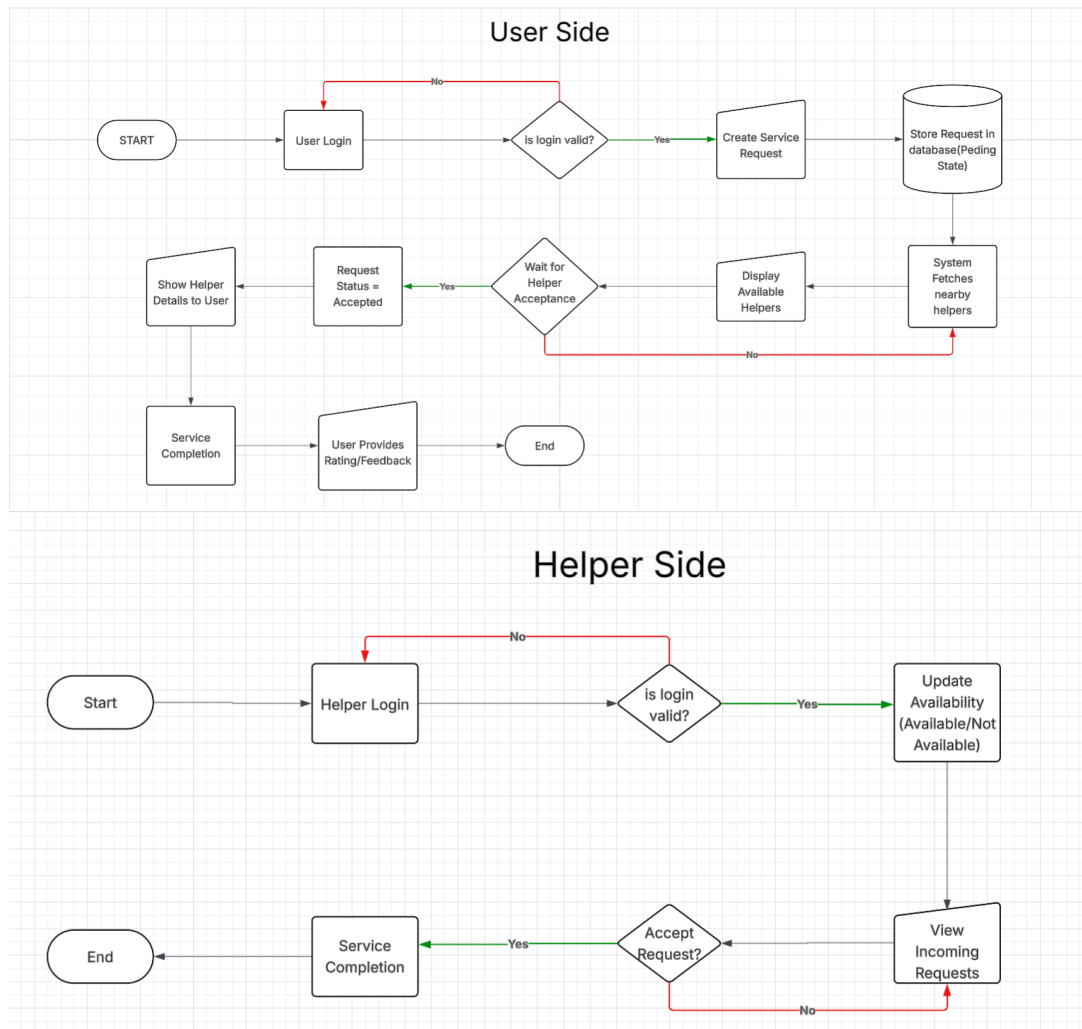


Figure 1: Flowchart of the Roadside Assistance System

4.4 Use Case Diagram

The use case diagram represents the interactions between the user, helper, and system in the Road Assistance System

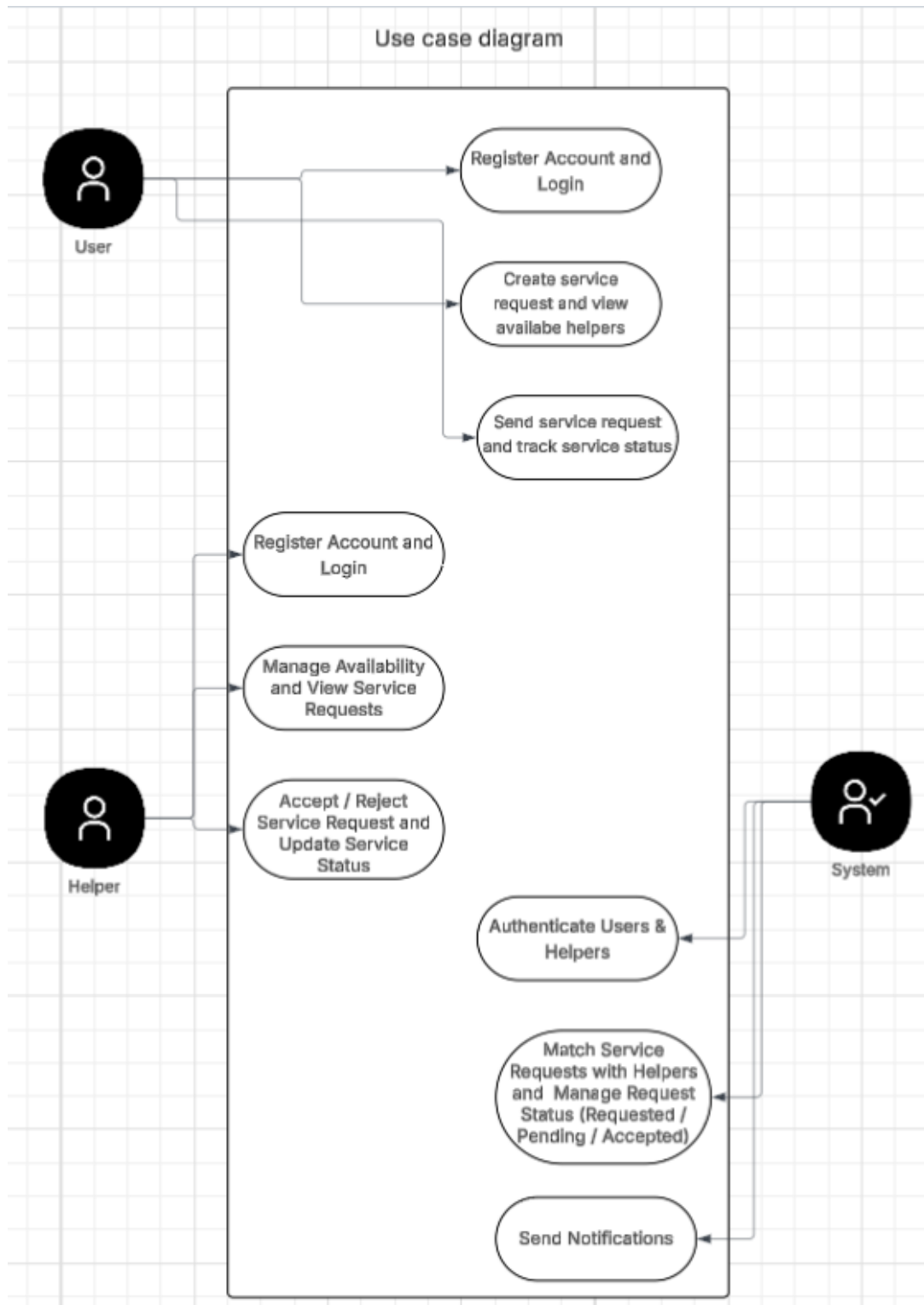


Figure 2: Use Case diagram of the Roadside Assistance System

4.5 Gantt Chart

The Gantt chart outlines the project timeline, showing key tasks, their start and end dates, and progress to track the development of the Road Assistance System.

